

NOISEXCAPE

URBAN NOISE PREDICTION WITH MACHINE LEARNING



TYPE: Machine Learning

YEAR: 2026

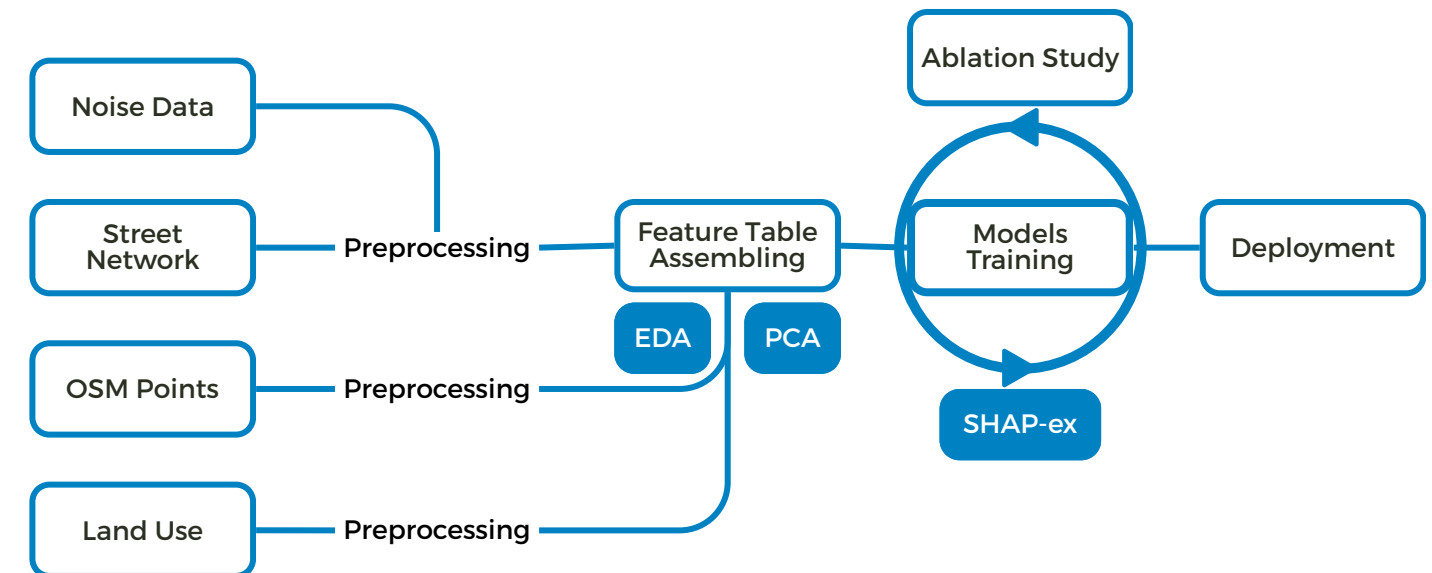
TEAM: Andrea Cutroni, Elena Corio, Symon Kipkemei, Giovanni Carlo Volpe

FACULTY: Gabriella Rossi and Georgios Bekakos

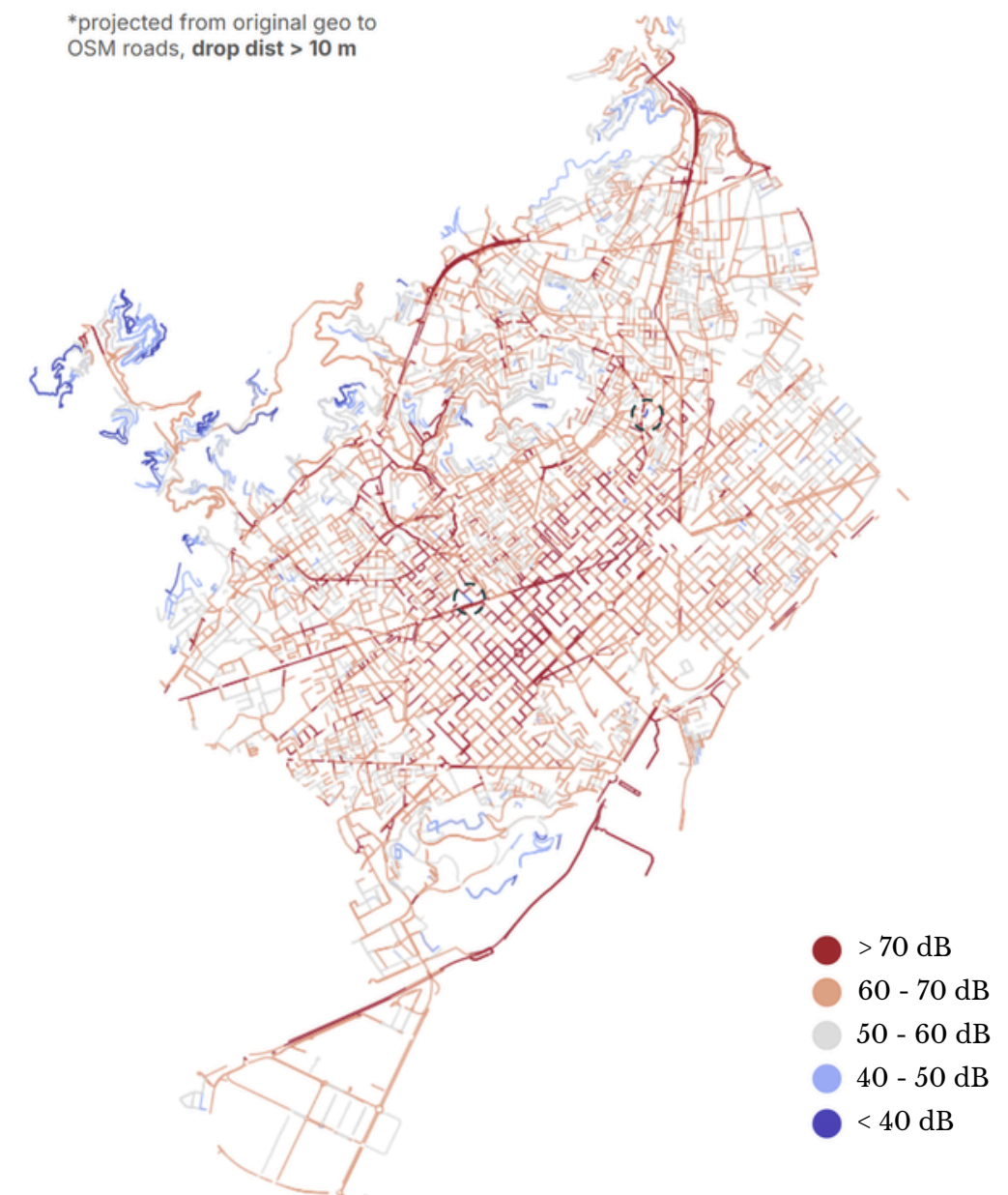
CONTEXT: IAAC Master in Advanced Computational Design (MACAD)

NoiseXcape explores whether urban noise can be predicted using open spatial data instead of costly traditional mapping. Using Barcelona data, 18 features - including land use, road proximity, space syntax centralities - were extracted from OpenStreetMap to train shallow learning models and an Artificial Neural Network (ANN). Principal Component Analysis (PCA) was used to reduce feature dimensionality, while SHAP analysis provided model interpretability. Tree-based models outperformed linear ones, with Random Forest achieving the best results. Cross-city testing showed that varying demographics affect transferability, though relative noise gradients remained accurate.

WORKFLOW



NOISE LEVELS



EDA

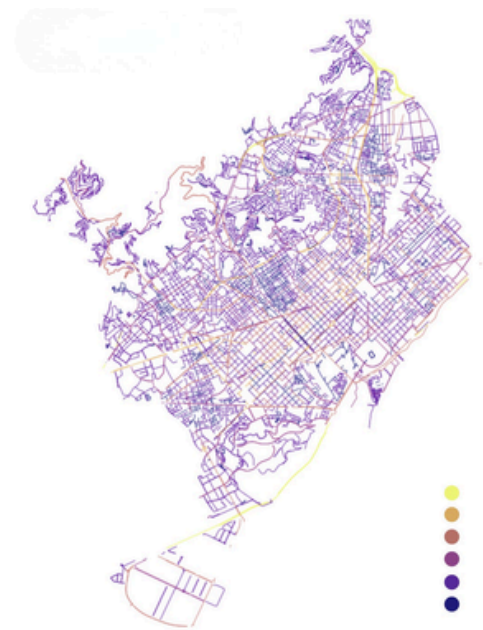
*colors in the maps are based on quantiles and not real scale



Betweenness Centrality

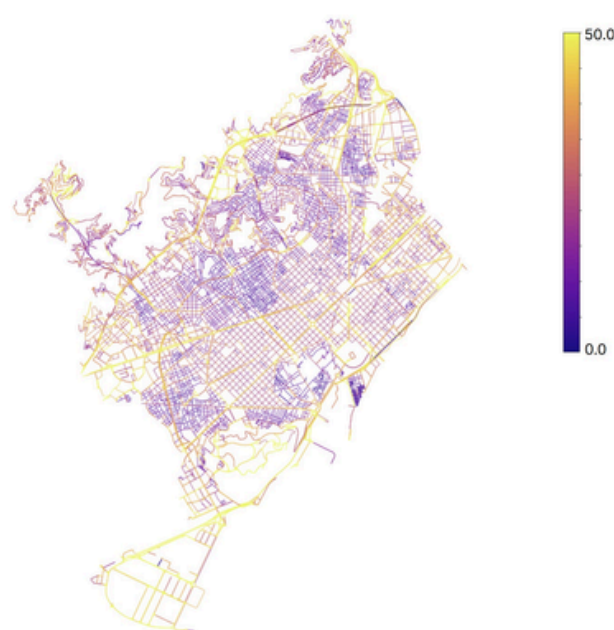


Straightness Centrality



Road Category

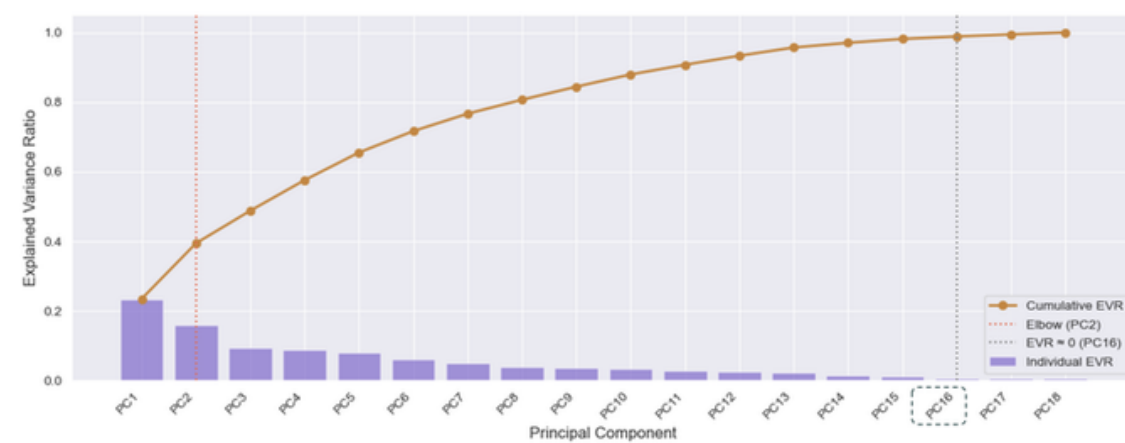
- 1 - Trunk
- 2 - Primary
- 3 - Secondary
- 4 - Tertiary
- 5 - Residential
- 6 - Living-street



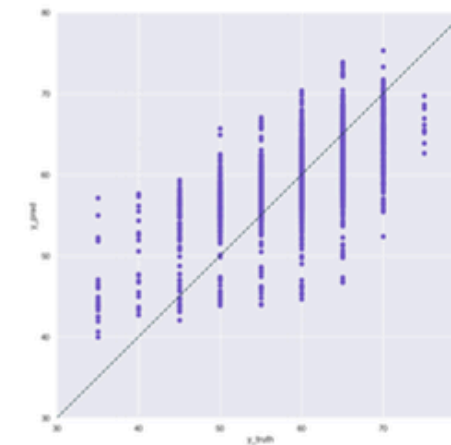
Total Street Width



PRINCIPAL COMPONENT ANALYSIS

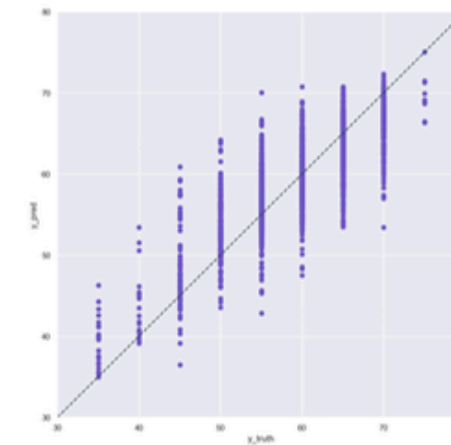


REGRESSION



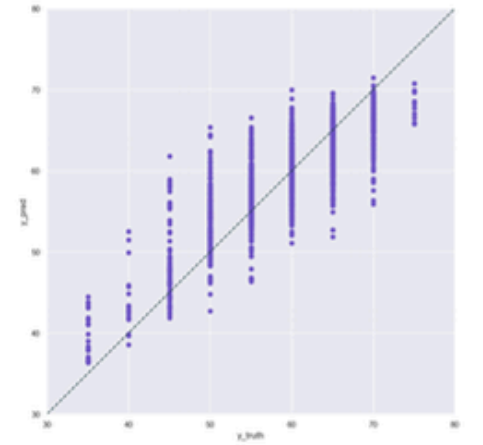
R^2 Train.: 0.455
 R^2 Test: 0.446

Linear Regression



R^2 Train.: 0.906
 R^2 Test: 0.675

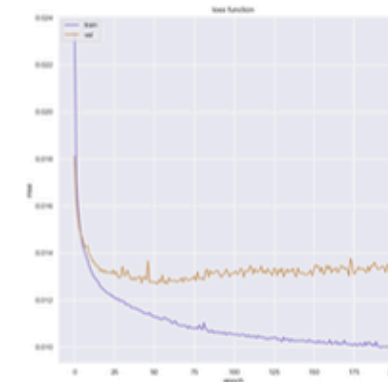
XGBoost



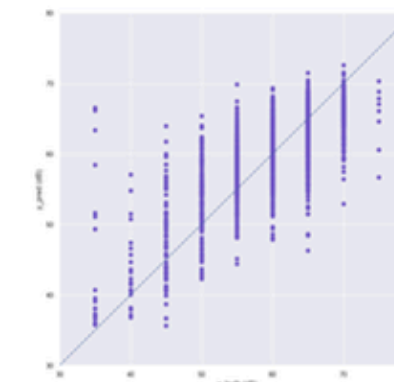
R^2 Train.: 0.959
 R^2 Test: 0.705

Random Forest

Loss Function



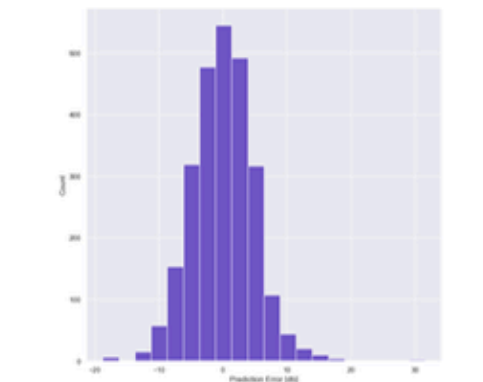
Prediction vs Truth



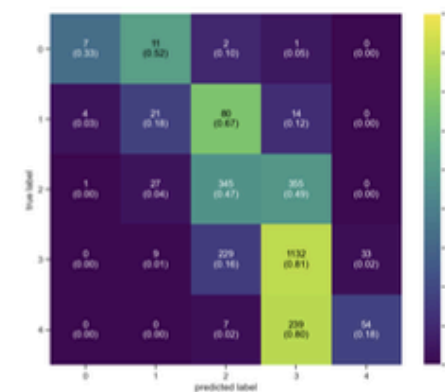
Test mse: 0.0143

Artificial Neural Network

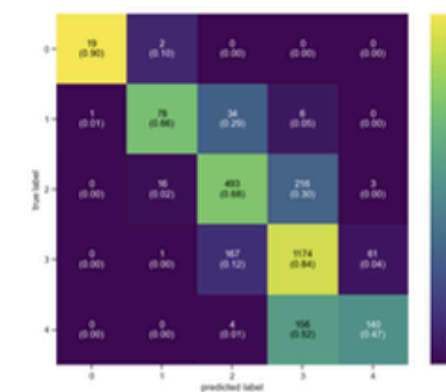
Prediction Error



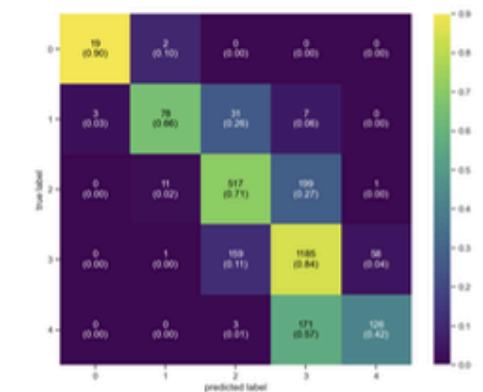
CLASSIFICATION



Logistic Regression



XGBoost



Random Forest